Python Programming (CST-005) - Unit 3

Unit 3 - Data Structures in Python - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: Python's four built-in data structures - list, tuple, set, dictionary - and strings.

Key ideas: list: ordered, mutable, allows duplicates - append(), insert(), remove(), pop(), sort(), slicing [a:b:c]. **tuple**: ordered but **immutable** - faster, hashable, used for fixed records. **set**: unordered, unique items - union (|), intersection (&), difference (-). **dict**: key → value pairs; keys must be immutable and unique - get(), keys(), values(), items(), update(). **Strings** are immutable sequences: slicing, upper()/lower(), split(), join(), find(), strip(). A **list comprehension** like [x*x for x in range(10) if x%2==0] builds a list in one line - examiners love it.

Exam pointers: "list vs tuple", "list vs set vs dict" comparisons and method-based programs (count word frequencies with a dict, remove duplicates with a set) are the repeated questions. Slicing output questions (negative indices) catch people - practise a few by hand.

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